

Homework II

Instructions

Show your code and the resulting output. Please ensure that your submission is clearly organized, readable, and easy to follow.

Question 1:

Generate a matrix of size $n \times n$ with the ij -th entry

$$\exp(-\theta(i-j)^2)$$

Specify $n = 10$ and $\theta = 0.5$.

```
theta <- 0.5
n <- 10
i <- matrix(rep(0:(n-1), each=n), n, n) # replicate from zero to 9
j <- matrix(rep(0:(n-1),n), n, n)
exp(-theta*(i-j)^2)
```

```
##           [,1]      [,2]      [,3]      [,4]      [,5]
## [1,] 1.000000e+00 6.065307e-01 1.353353e-01 1.110900e-02 3.354626e-04
## [2,] 6.065307e-01 1.000000e+00 6.065307e-01 1.353353e-01 1.110900e-02
## [3,] 1.353353e-01 6.065307e-01 1.000000e+00 6.065307e-01 1.353353e-01
## [4,] 1.110900e-02 1.353353e-01 6.065307e-01 1.000000e+00 6.065307e-01
## [5,] 3.354626e-04 1.110900e-02 1.353353e-01 6.065307e-01 1.000000e+00
## [6,] 3.726653e-06 3.354626e-04 1.110900e-02 1.353353e-01 6.065307e-01
## [7,] 1.522998e-08 3.726653e-06 3.354626e-04 1.110900e-02 1.353353e-01
## [8,] 2.289735e-11 1.522998e-08 3.726653e-06 3.354626e-04 1.110900e-02
## [9,] 1.266417e-14 2.289735e-11 1.522998e-08 3.726653e-06 3.354626e-04
## [10,] 2.576757e-18 1.266417e-14 2.289735e-11 1.522998e-08 3.726653e-06
##           [,6]      [,7]      [,8]      [,9]      [,10]
## [1,] 3.726653e-06 1.522998e-08 2.289735e-11 1.266417e-14 2.576757e-18
## [2,] 3.354626e-04 3.726653e-06 1.522998e-08 2.289735e-11 1.266417e-14
## [3,] 1.110900e-02 3.354626e-04 3.726653e-06 1.522998e-08 2.289735e-11
## [4,] 1.353353e-01 1.110900e-02 3.354626e-04 3.726653e-06 1.522998e-08
## [5,] 6.065307e-01 1.353353e-01 1.110900e-02 3.354626e-04 3.726653e-06
## [6,] 1.000000e+00 6.065307e-01 1.353353e-01 1.110900e-02 3.354626e-04
## [7,] 6.065307e-01 1.000000e+00 6.065307e-01 1.353353e-01 1.110900e-02
## [8,] 1.353353e-01 6.065307e-01 1.000000e+00 6.065307e-01 1.353353e-01
## [9,] 1.110900e-02 1.353353e-01 6.065307e-01 1.000000e+00 6.065307e-01
## [10,] 3.354626e-04 1.110900e-02 1.353353e-01 6.065307e-01 1.000000e+00
```

Question 2:

Use for loop and if statement to develop a function that can sort a vector of entries from the smallest to the largest. For example, you can implement the bubble sort algorithm. Bubble sort repeatedly looks at two neighboring items.

Bubble sort compares neighboring items from left to right. If the item on the left is larger than the item on the right, the two items are swapped. After the first pass, the largest unsorted item moves to the end. During the second pass, the algorithm starts again from the left but stops before the last sorted item. It repeats this process until no swaps are needed.

The algorithm repeats this process until it completes a pass without making any swaps. At that point, the items are in ascending order.

For example, To sort 3, 7, 0, -5 in ascending order:

First pass

Compare 3 and 7: no swap.

Compare 7 and 0: swap and result 3, 0, 7, -5

Compare 7 and -5: swap and result 3, 0, -5, 7

Second pass

Compare 3 and 0: swap and result 0, 3, -5, 7

Compare 3 and -5: swap and result 0, -5, 3, 7

Third pass

Compare 0 and -5: swap and result -5, 0, 3, 7 Compare 0 and 3: no swap.

The sorted vector is:

-5, 0, 3, 7

Use your function to order the following vectors

```
## this is a function that take a vector v
bubble_sort <- function(v)
{ # look at bubble sort algorithm on youtube
  # calculate the length of array
  n <- length(v)
  # run loop n-1 times
  for (i in 1 : (n - 1)) {
    # run loop (n-i) times
    for (j in 1 : (n - i)) {
      # compare elements
      if (v[j] > v[j + 1]) {
        update_val <- v[j]
        v[j] <- v[j + 1]
        v[j + 1] <- update_val
      }
    }
  }
  v
}
```

x <- c(5, 8, 9, 3, 2)

```

y <- c(4, 1, 8, 0, -3)

bbs_x <- bubble_sort(x)
bbs_y <- bubble_sort(y)
cat("The sorted vector for x is", bbs_x)

## The sorted vector for x is 2 3 5 8 9

cat("\nThe sorted vector for y is ", bbs_y)

##
## The sorted vector for y is  -3 0 1 4 8

```

Question 3

Use the “Bschoool.csv” dataset to find the answers of the following question within this dataset.

1. Which professor rank (i.e., associate, assistant, or professor) has the highest average teaching evaluation score?

```

Bschoool <- read.csv("C:/Users/FRT/Downloads/Fall-2026/MATH-8830/Bschoool.csv")
mean_score <- tapply(Bschoool$score, Bschoool$title, mean)
highest <- max(mean_score)
print(mean_score)

```

```

##           Professor  AssistantProfessor Associate Professor
##           17.57574           16.50972           16.75487

```

```

cat("The Professor has the higher rank with", highest)

```

```

## The Professor has the higher rank with 17.57574

```

2. What is the number of female associate professors?

```

female_associate_professors <- sum(Bschoool$title == "Associate Professor" &
                                   Bschoool$gender == "f")

cat("\n The number of female associates professor:", female_associate_professors)

```

```

##
## The number of female associates professor: 21

```

3. Is median teaching evaluation score in fall semester usually higher than that in spring semester?

```

eva_fall <- median(Bschoool$score[Bschoool$semester == "fall"])
eva_spring <- median(Bschoool$score[Bschoool$semester == "spring"])
eva_fall >= eva_spring # this return FALSE, so the answer is NO

```

```

## [1] FALSE

```

```
#cat("\n The median score that is high is in the spring", eva_spring)
```